CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 01
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a Java program to calculate calorie intake during Iftar. The program should ask the user to enter a six-digit number, and the product of the last three digits of this number will be used as the total calorie limit. The user must then input how many dates they want to eat and how many jilapis they want to eat. Each date contains 80 calories and each jilapi contains 140 calories, and the person must always eat dates first. The program should first calculate the total calories from the dates; if the calories from the dates exceed the calorie limit, then the person cannot eat any jilapis and must drink water instead. If the calories from the dates are within the limit, the program should calculate how many jilapis can be eaten without exceeding the total calorie limit.

Input	Output
Enter a 6-digit number: 123456 Enter number of dates: 2 Enter number of jilapis: 3	Calorie limit: 120 Calories from dates: 160 Dates exceed the calorie limit. You cannot eat jilapis. Drink water instead.
Enter a 6-digit number: 243798 Enter number of dates: 2 Enter number of jilapis: 3	Calorie limit: 504 Calories from dates: 160 You can eat up to 2 jilapis. Total calories consumed: 440

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 01
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a **Java program** that calculates calorie intake during Sehri. The program should take input for the number of rice plates, glasses of milk, and bananas a person wants to eat. Each plate of rice has 200 calories, each glass of milk has 120 calories, and each banana has 100 calories, with a total calorie limit of 800. The program must calculate rice calories first, and if rice alone exceeds 800 calories, the person can only drink water. If not, then calculate the calories from milk and bananas and make sure the total does not go over 800 calories.

Sample Input	Sample Output
Enter number of plates of rice: 5 Enter number of glasses of milk: 1 Enter number of bananas: 1	Rice calories = 1000 Calorie limit exceeded by rice alone! You should drink only water.
Enter number of plates of rice: 2 Enter number of glasses of milk: 1 Enter number of bananas: 1	Rice calories = 400 Milk calories = 120 Banana calories = 100 Total calorie intake = 620 You are within the 800 calorie limit. Good Sehri!
Enter number of plates of rice: 3 Enter number of glasses of milk: 2 Enter number of bananas: 2	Rice calories = 600 Adding milk and bananas exceeds the 800 calorie limit. You should reduce milk or bananas. Maximum remaining calories after rice = 200

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A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a Java program to calculate calorie intake during Iftar. The program should ask the user to enter a six-digit number, and the product of the last three digits of this number will be used as the total calorie limit. The user must then input how many dates they want to eat and how many jilapis they want to eat. Each date contains 80 calories and each jilapi contains 140 calories, and the person must always eat dates first. The program should first calculate the total calories from the dates; if the calories from the dates exceed the calorie limit, then the person cannot eat any jilapis and must drink water instead. If the calories from the dates are within the limit, the program should calculate how many jilapis can be eaten without exceeding the total calorie limit.

Input	Output
Enter a 6-digit number: 123456 Enter number of dates: 2 Enter number of jilapis: 3	Calorie limit: 120 Calories from dates: 160 Dates exceed the calorie limit. You cannot eat jilapis. Drink water instead.
Enter a 6-digit number: 243798 Enter number of dates: 2 Enter number of jilapis: 3	Calorie limit: 504 Calories from dates: 160 You can eat up to 2 jilapis. Total calories consumed: 440

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B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a **Java program** that calculates calorie intake during Sehri. The program should take input for the number of rice plates, glasses of milk, and bananas a person wants to eat. Each plate of rice has 200 calories, each glass of milk has 120 calories, and each banana has 100 calories, with a total calorie limit of 800. The program must calculate rice calories first, and if rice alone exceeds 800 calories, the person can only drink water. If not, then calculate the calories from milk and bananas and make sure the total does not go over 800 calories.

Sample Input	Sample Output
Enter number of plates of rice: 5 Enter number of glasses of milk: 1 Enter number of bananas: 1	Rice calories = 1000 Calorie limit exceeded by rice alone! You should drink only water.
Enter number of plates of rice: 2 Enter number of glasses of milk: 1 Enter number of bananas: 1	Rice calories = 400 Milk calories = 120 Banana calories = 100 Total calorie intake = 620 You are within the 800 calorie limit. Good Sehri!
Enter number of plates of rice: 3 Enter number of glasses of milk: 2 Enter number of bananas: 2	Rice calories = 600 Adding milk and bananas exceeds the 800 calorie limit. You should reduce milk or bananas. Maximum remaining calories after rice = 200

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 02
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a Java program to analyze food sales during the celebration of Pahela Baishakh at BRAC University. The program should ask the user to enter the number of student stalls in the festival. For each student stall, the number of items sold is equal to the stall number. For example, Stall 1 sells 1 item, Stall 2 sells 2 items, Stall 3 sells 3 items, and so on. For each item, the user must input its price.

For every student stall, the program should display all items that cost more than 200 taka (premium items) and calculate the total sales of that stall. If a stall has 2 or more premium items, the program should print "**Popular Stall!**". After processing all stalls, the program should display the overall total sales of the festival.

Input	Output
Enter number of stalls: 2 Stall 1: Enter price of item 1: 220 Stall 2: Enter price of item 1: 150 Enter price of item 2: 250	Stall 1: Premium items: 220 Total sales: 220 Stall 2: Premium items: 250 Total sales: 400 Overall Festival Sales: 620
Enter number of stalls: 3 Stall 1: Enter price of item 1: 150 Stall 2: Enter price of item 1: 250 Enter price of item 2: 300 Stall 3: Enter price of item 1: 100 Enter price of item 2: 220 Enter price of item 3: 50	Stall 1: Premium items: Total sales: 150 Stall 2: Premium items: 250 300 Total sales: 550 **Popular Stall!** Stall 3: Premium items: 220 Total sales: 370 Overall Festival Sales: 1070

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 02
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Write a Java program to analyze cultural sessions during the celebration of Pahela Baishakh at BRAC University. The program should ask the user to enter the number of sessions. For each session, the program should first take input of the number of performances. Then, for each performance, the user must input its duration (in minutes).

For every session, the program should count how many performances have a duration greater than 30 minutes and calculate the total duration of that session. If more than half of the performances in a session have a duration greater than 30 minutes, the program should print "**Long Session!**". After processing all sessions, the program should display the total duration of all sessions combined.

Input	Output
Enter number of sessions: 1 Session 1: Enter number of performances: 4 Enter duration of performance 1: 10 Enter duration of performance 2: 35 Enter duration of performance 3: 45 Enter duration of performance 4: 20	Session 1: Long performances count: 2 Total duration: 110 Overall Total Duration: 110
Enter number of sessions: 2 Session 1: Enter number of performances: 3 Enter duration of performance 1: 20 Enter duration of performance 2: 40 Enter duration of performance 3: 50 Session 2: Enter number of performances: 2 Enter duration of performance 1: 25 Enter duration of performance 2: 30	Session 1: Long performances count: 2 Total duration: 110 **Long Session!** Session 2: Long performances count: 0 Total duration: 55 Overall Total Duration: 165

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Lab Exam - 02
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

The water park needs a program to manage visitor eligibility for various activities and special offers. Write a Java program that first takes the number of visitor sessions, followed by the number of visitors in each session. For each visitor, the program should further take the Activity Level (1 for Low, 2 for Moderate, 3 for High) and whether the visitor has visited the park previously (true for previous visitors, false for first-time visitors). The park offers the following activities and special offers: the **Lazy Pool** is available for visitors with an activity level of 2 or 3; the **Water Coaster** is available for visitors with an activity level of 3 who have visited the park before; and the **Special Offer** is available to first-time visitors. The program should count and display the total number of eligible visitors for each activity and for the Special Offer.

Sample Input	Sample Output
Enter number of visitor sessions: 2 Enter number of visitors in session 1: 3 Enter activity level of visitor 1: 2 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 1 Enter previous visit of visitor 2: false Enter activity level of visitor 3: 2 Enter previous visit of visitor 3: false Enter number of visitors in session 2: 2 Enter activity level of visitor 1: 3 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 2 Enter previous visit of visitor 2: false	Lazy Pool Riders: 4 Water Coaster Riders: 1 Visitors with Special Offer: 3 **Explanation:** For session 1, Visitor 1 qualifies for the Lazy Pool but not the Water Coaster due to an activity level of 2. Visitor 2 qualifies for the Special Offer as a first-time visitor. Visitor 3 qualifies for the Lazy Pool and Special Offer. For session 2, Visitor 1 qualifies for both the Lazy Pool and the Water Coaster. Visitor 2 qualifies for the Lazy Pool and the Special Offer.

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Spring 2026
Lab Exam - 02
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

You are asked to write a Java program for an amusement park that manages visitor eligibility for multiple rides and special offers. The program first takes the number of entry sessions from the management team. For each session, the program takes the number of visitors, followed by each visitor's height (in cm) and age (in years). The park has multiple rides with eligibility rules: the **Roller Coaster** requires a minimum height of 140 cm and a minimum age of 12 years, the **Ferris Wheel** allows visitors aged 5 years or older, and the **Bumper Cars** allow visitors aged 8 years or older with a maximum height of 200 cm for safety. Additionally, visitors under 12 years are eligible for a **Special Discount Offer**. The program should count the number of visitors who are eligible for each ride, as well as how many receive the discount. Finally, the program should display these counts after processing all sessions.

Sample Input	Sample Output
Enter number of sessions = 3 Enter number of visitors in session 1 = 3 Enter height of visitor 1: 150 Enter age of visitor 1: 14 Enter height of visitor 2: 120 Enter age of visitor 2: 10 Enter height of visitor 3: 160 Enter age of visitor 3: 20 Enter number of visitors in session 2 = 2 Enter height of visitor 1: 130 Enter age of visitor 1: 6 Enter height of visitor 2: 170 Enter age of visitor 2: 13 Enter number of visitors in session 3 = 1 Enter height of visitor 1: 180 Enter age of visitor 1: 9	Roller Coaster Riders: 3 Ferris Wheel Riders: 6 Bumper Cars Riders: 5 Visitors with Special Offer: 3 **Explanation:** Visitor 1 in session 1 qualifies for all rides, Visitor 2 qualifies for the Ferris Wheel and Bumper Cars and gets the Special Offer, and Visitor 3 qualifies for all rides. In session 2, Visitor 1 qualifies for the Ferris Wheel and the Special Offer, while Visitor 2 qualifies for all rides. In session 3, the visitor qualifies for the Ferris Wheel and Bumper Cars and receives the Special Offer.

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Spring 2025
Lab Exam - 03
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

Write a Java program that reads a string from the user and produces its run-length encoded form. In run-length encoding, each group of consecutive identical characters is replaced by the character followed by its count. If a character appears only once in a row, write the character without a number. After producing the encoded string, compare its length with the original. If the encoded version is shorter, print the encoded string. Otherwise, print the original string unchanged.

Sample Input	Sample Output
aabbbccddddee	a2b3c2d4e2
abcd	abcd (encoded "a1b1c1d1" is longer, so print original)
aaabbaaa	a3b2a3

CSE110: Programming Language I (Lab)
Spring 2025
Lab Exam - 03
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

Write a Java program that reads a main string and a pattern string, then prints how many times the pattern appears in the main string (non-overlapping occurrences only). You must implement the search manually, no built-in method may be used to find or detect the substring.

Sample Input	Sample Output
main: ababab pattern: ab	Count: 3
main: aaaaaa pattern: aa	Count: 3
main: hello world pattern: xyz	Count: 0

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 03
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (* or /) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q *	371 531 197001
5dQ!87 26f0&3 /	57 3 19
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (*), the final output is 197001	

CSE110: Programming Language I (Lab)
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Lab Exam - 03
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (+ or -) from the user. Your task is to create two numbers from the two input strings by considering only the even digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q +	462 480 942
2dQ!83 36f1&5 -	28 6 22
Explanation: For sample Input 1, ab4s3F67h%21 contains even digits 4,6,2, that's why the first number is 462 mnP543&1b86Q contains even digits 4,8,0, that's why the second number is 480 After performing the addition operation (+), the final output is 942	

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 04
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

A weekly attendance system stores 7 digits in an array, where **each index** represents a day of the week in order from Saturday to Friday. In this system, classes on **Saturday** (index 0) and **Tuesday** (index 3) are conducted **online**, while all other days are conducted **in-person**. A day is considered balanced if the number of elements before it (to the left) with lower values is equal to the number of elements after it (to the right) with higher values.

Write a Java program to determine and print **all** balanced days **only among the in-person days**, and also print the **total** number of such balanced in-person days.

Given Array	**Output**
int[] arr = {40, 10, 20, 100, 15, 30, 5};	Balanced in-person days are: 2 4 6 Total balanced in-person days: 3
Explanation: Monday (index **2**, value 20): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Wednesday (index **4**, value 15): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Friday (index **6**, value 5): left lower count = 0, right higher count = 0, ignoring indices 0 and 3 (online). Since both are equal, the day is balanced.	

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CSE110: Programming Language I (Lab)
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B

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

A FIFA World Cup performance tracker stores the number of goal chances created for 7 matches in an array, where **each index** represents a match in sequence from Match 1 to Match 7. In this system, matches played on **Monday** (index 1) and **Thursday** (index 4) are against **rival teams**, while all other matches are **non-rival.** A match is considered Growth-Stable if the number of non-rival matches before it (to the left) with higher values is equal to the number of non-rival matches after it (to the right) with lower values.

Write a Java program to determine and print **all** Growth-Stable matches **only among the non-rival matches**, and also print the **total** number of such Growth-Stable matches.

<table>
<tr><th>Given Array</th><th>Output</th></tr>
<tr><td>int[] scores = {40, 10, 20, 25, 15, 30, 5};</td><td>Growth-Stable matches are: 2 3 5

Total Growth-Stable matches: 3</td></tr>
<tr><td colspan="2">Explanation:
Match 3 (index 2, value 20): left higher count = 1 (40), right lower count = 1 (5), ignoring indices 1 and 4. Since both are equal, the match is Growth-Stable.

Match 4 (index 3, value 25): left higher count = 1 (40), right lower count = 1 (5), ignoring indices 1 and 4. Since both are equal, the match is Growth-Stable.

Match 6 (index 5, value 30): left higher count = 1 (40), right lower count = 1 (5), ignoring indices 1 and 4. Since both are equal, the match is Growth-Stable.</td></tr>
</table>

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 4
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

Write a Java program that rearranges the elements of a given array in such a way that all even numbers appear before all odd numbers. *[**Do it in-place. You can't create a new array to solve it. The relative order among even or odd numbers does not matter.**] [**Your code should work for any given integer array**]*

Given Array	Expected Output
int [] arr = { 3, 6, 4, 7, 1, 9, 2};	Rearranged Array: 6, 4, 2, 7, 1, 9, 3
int[] arr = { 2, 11, 1, 7, 12, 8, 3};	Rearranged Array: 2, 12, 8, 7, 11, 1, 3

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Spring 2026
Lab Exam - 4
Duration: 25 Minutes

B

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

Write a Java program that rearranges the elements of a given array in such a way that all odd numbers appear before all even numbers. *[**Do it in-place. You can't create a new array to solve it. The relative order among even or odd numbers does not matter.**] [**Your code should work for any given integer array**]*

Given Array	Expected Output
int [] arr = {5, 4, 2, 7, 8, 9};	Rearranged Array: 5, 7, 9, 4, 8, 2
int[] arr = { 4, 16, 14, 3, 7, 12, 5};	Rearranged Array: 3, 7, 5, 4, 16, 12, 14